Advik Rajvansh

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Portfolio: advik-rajvansh.vercel.app

Summary

Third-year B.Tech CSE student specializing in **Machine Learning and Data Science**. Experienced in building and deploying end-to-end ML pipelines in Python — from EDA and feature engineering to model training, evaluation, and Flask deployment — with hands-on projects reaching **88%+ predictive accuracy**. Strong foundation in Data Structures, OOP, and SQL, reinforced by **167+ solved LeetCode problems**. Smart India Hackathon college-level qualifier passionate about turning data into actionable insight.

Education

JIIT Sector 62, Noida, UP — *July 2024 – July 2028 (Expected)*
Bachelor of Technology (B.Tech) | Major: Computer Science — **CGPA: 7.6/10.0**
Relevant Coursework: Data Structures & Algorithms, OOP in C++, DBMS, Operating Systems, Machine Learning Basics

Technical Skills

- **Programming Languages:** Python, C++, SQL, Java, Bash/Unix
- **ML & Data Science:** Scikit-Learn, XGBoost, Pandas, NumPy, SMOTE, Cross-Validation, Feature Engineering
- **ML Techniques:** Classification, Regression, Clustering, TF-IDF & BM25 Information Retrieval, Model Evaluation (Accuracy, F1, MAE, R^2)
- **Data & Visualization:** EDA, Data Cleaning, Preprocessing, Matplotlib, Seaborn
- **Web & Deployment:** Flask, REST APIs, HTML/CSS
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems
- **Tools:** Jupyter Notebook, Git/GitHub, Google Colab

Projects

Heart Disease & Cardiovascular Risk Prediction — *2026*
Data Science / Machine Learning Project
- Built an end-to-end machine learning pipeline to predict cardiovascular risk on **8,700+ patient** clinical and lifestyle records.
- Engineered domain-driven features (Mean Arterial Pressure, composite Lifestyle Score, Age–Cholesterol Index) and performed univariate, bivariate, and multivariate EDA.
- Trained and benchmarked **Logistic Regression, Random Forest, and XGBoost**, achieving **88.5% accuracy and a 0.877 F1-score**, validated with Stratified K-Fold cross-validation.
- Addressed class imbalance using **SMOTE** oversampling and decision-threshold optimization to improve high-risk recall.
- *Tools: Python, Pandas, NumPy, Scikit-Learn, XGBoost, Matplotlib, Seaborn.*
- Repository: github.com/lucifier950/Heart-Disease-Heart-Attack-Risk-Prediction

MANN-O-METER — *Nov 2025 – Dec 2025*
Machine Learning + Flask Project
- Built an AI-based mental health support system using **Flask**, integrating a chatbot with ML-based stress prediction.
- Implemented an information retrieval chatbot using **TF-IDF + BM25** to answer mental health FAQ queries.
- Developed a stress analyzer using **RandomForestRegressor** and **RandomForestClassifier** for regression and classification.
- Performed feature engineering including gender encoding, blood pressure parsing, and derived health-metric creation.
- Evaluated model performance using MAE, R^2, Accuracy, Confusion Matrix, and Classification Report.
- Repository: github.com/lucifier950/Mann-o-meter

Workshops Attended

- Deep Learning Workshop — **IIT Delhi**
- Retrieval-Augmented Generation (RAG) & Large Language Models (LLMs) Workshop — **JIIT**

Achievements & Leadership

- Selected at the college level for the **Smart India Hackathon (SIH)**, competing among top student teams.
- Solved **167+ C++ problems on LeetCode** (leetcode.com/u/lucifier121), strengthening problem-solving skills and DSA fundamentals.